while
statements

CSCI 134

back

let's go to

vegas

the rules of craps

phase one:

already implemented

- the player rolls two standard dice
- if the total is 7 or 11, then the player wins
- if the total is 2, 3, or 12, then the player loses
- otherwise, we refer to the total as the "point" and proceed to phase two

phase two

- the player continues to roll until the dice is 7 (in which case, the player loses) or until the dice total equals the point (in which case, the player wins)

phase one: *already implemented*

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- if the total is 2, 3, or 12, then the player loses
- otherwise, we refer to the total as the "point" and proceed to phase two

```python
from random import randint

def roll_two_dice():
    die1 = randint(1, 6)  # roll the first die
    die2 = randint(1, 6)  # roll the second die
    dice_total = die1 + die2
    print("You rolled a " + str(dice_total))
    return dice_total

def phase_one():
    point = None
    total = roll_two_dice()
    if total == 7 or total == 11:
        print("... you win!")
    elif total == 2 or total == 3 or total == 12:
        print("... you lose!")
    else:
        point = total
        print("  proceed to phase two!")
    return point
```

phase two

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    else:
        point = total
        print("  proceed to phase two!")
    return point

def phase_two(point):
    # what should we do?
```

idea 1: use if-statements

the player continues to roll until the dice total equals 7 or the point

```
def phase_two(point):
    total = roll_two_dice()
    if total != 7 and total != point:
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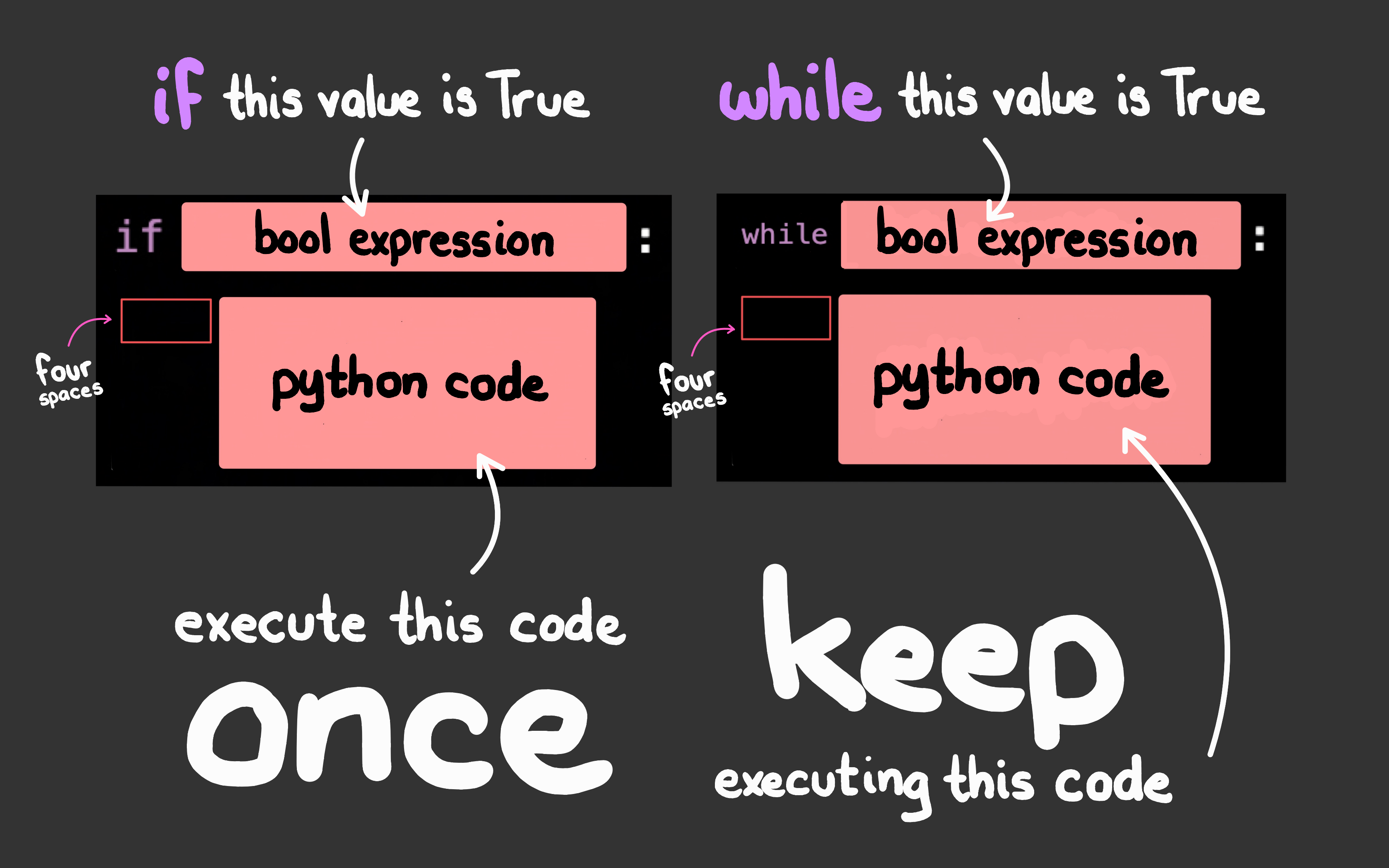
    if total != 7 and total != point:
        total = roll_two_dice()
    ...
```

but we can't keep writing if-statements forever

alternative: while statements

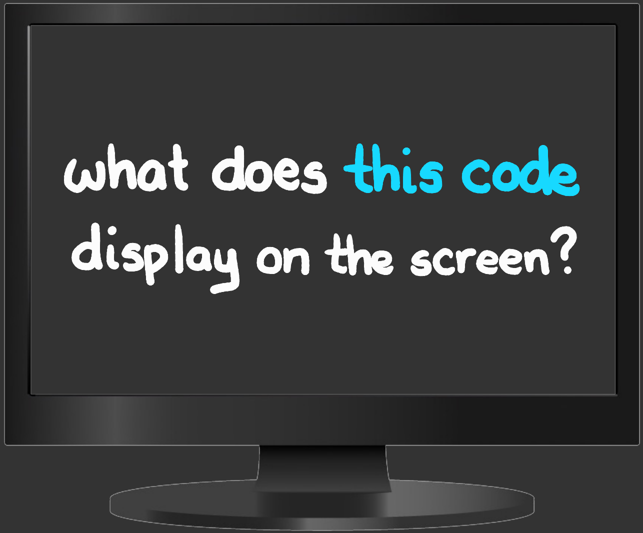
```
def phase_two(point):
    total = roll_two_dice()
    while total != 7 and total != point:
        total = roll_two_dice()
```

the player continues to roll until the dice total equals 7 or the point



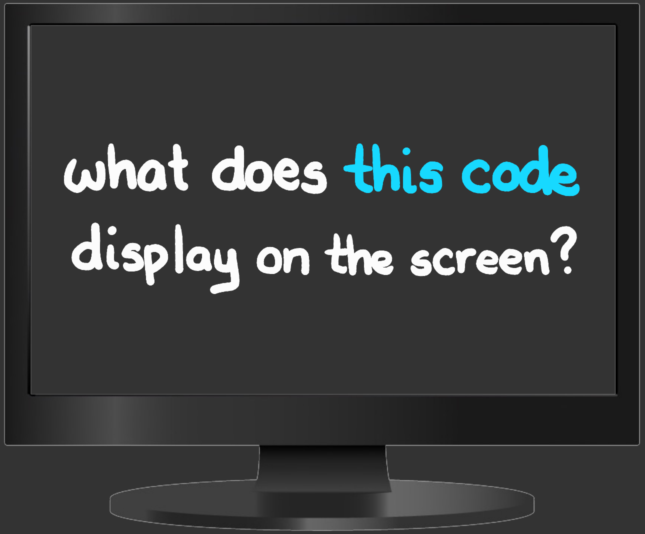
if this value is True
if
bool expression
:
four spaces
python code
execute this code
once
while this value is True
while
bool expression
:
four spaces
python code
keep
executing this code

```
countdown = 10
while countdown > 0:
    print(countdown)
    countdown = countdown - 1
print('blastoff')
```

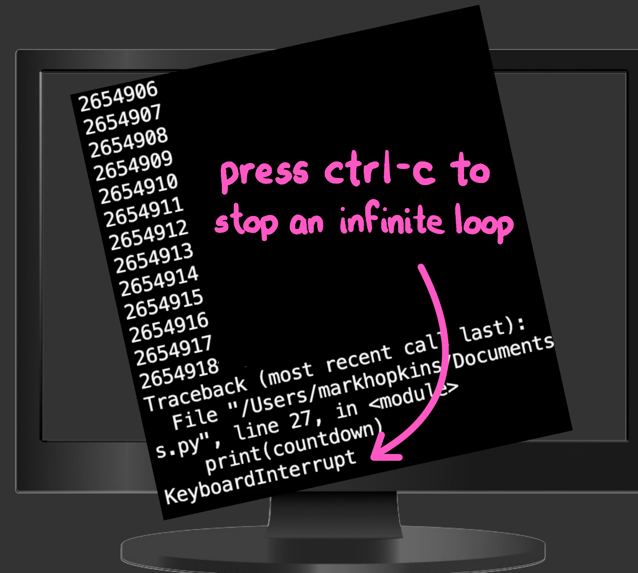


```python
countdown = 10
while countdown > 0:
    print(countdown)
    countdown = countdown - 1
print('blastoff')
```

```python
countdown = 10
while countdown > 0:
    print(countdown)
    countdown = countdown + 1
print('blastoff')
```



```python
countdown = 10
while countdown > 0:
    print(countdown)
    countdown = countdown + 1
print('blastoff')
```



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def phase_one():
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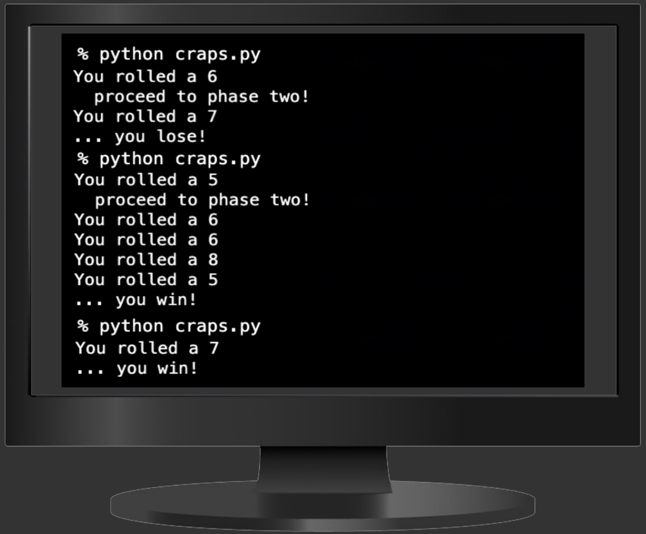
    elif total == 2 or total == 3 or total == 12:
        print("... you lose!")
    else:
        point = total
        print("  proceed to phase two!")
    return point

def phase_two(point):
    total = roll_two_dice()
    while total != 7 and total != point:
        total = roll_two_dice()
    if total == point:
        print("... you win!")
    else:
        print("... you lose!")

point = phase_one()
if point != None:
    phase_two(point)
```



a complete craps game in python

statements so far

assignment

variable = expression

def

def name (arguments (comma-separated)):
4 spaces
return return value (what the function computes)

for

for variable in range(int , int):
python code

if

if bool expression :
python code

while

while bool expression :
python code